



UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2020/2021 – First Year Examination – Semester II – 2021

SCS 1211 – Mathematical Methods I

TWO (2) HOURS



057

Important Instructions to Candidates:

1. The medium of instructions and questions is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the answer paper.
5. This paper has **4** questions and **4** pages.
6. Answer **ALL** questions. All questions carry equal marks (**25** marks).
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
8. **Non-Programmable/Programmable** calculators are **not allowed**.

1.

- (a) Let $n \in \mathbb{Z}^+$. Give an example of two vector spaces V and W over the fields E and F respectively such that $V = W$, $\dim V = n$, and $\dim W = 2n$. Justify your answer.

[12 Marks]

- (b) Let $n \in \mathbb{Z}^+$ be such that $n > 1$.

$$\text{Let } S = \left\{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : \begin{vmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-1} & x_n \\ 0 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 0 & 1 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{vmatrix} = 0 \right\}.$$

- (i) Prove that S is a subspace of the Euclidean space $\mathbb{R}^n$.

- (ii) Prove that $S = \text{Span}\{(0, 1, 1, \dots, 1, 1), (0, 0, 1, \dots, 1, 1), \dots, (0, 0, 0, \dots, 0, 1)\}$.

[8 Marks]

- (c) Let $n \in \mathbb{Z}^+$ be such that $n > 1$ and let $V = \{0, 1, 2, \dots, n-1\}$. Let $\oplus$ be the associative binary operation defined on V in the following way:

Let $u, w \in V$. Then, $u \oplus w = v$, where $v \in V$ is the non-negative remainder obtained when $u + w$ is divided by n .

Does there exist a scalar multiplication $\odot: \mathbb{R} \times V \rightarrow V$ such that V together with $\oplus$ and $\odot$ is a vector space over the field $\mathbb{R}$? Justify your answer.

[5 Marks]

2.

- (a) Let V be a vector space over the field F and let $T, U: V \rightarrow V$ be two linear transformations.

- (i) Prove that $U \circ T: V \rightarrow V$, defined for each $v \in V$ by $(U \circ T)(v) = U(T(v))$, is a linear transformation.

- (ii) Let $\mathbb{R}^3$ be the Euclidean space over $\mathbb{R}$. Prove that there exists $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^3)$ such that $T \neq I$, $T^2 \neq I$, and $T^{2022} = I$, where $I: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the identity transformation.

[9 Marks]

- (b) Let $S = \{e_1, e_2, e_3\}$ be the standard ordered basis for $\mathbb{R}^3$, where $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, and $e_3 = (0, 0, 1)$. Consider the linear transformations $T, U: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T((x, y, z)) = (x - y - z, -x + y - z, -x - y + z)$ and $U((x, y, z)) = (-x + y + z, x - y + z, x + y - z)$ for each $(x, y, z) \in \mathbb{R}^3$. Compute $[T]_{SS}$ and $[U]_{SS}$.

[8 Marks]

(c) Find $[U \circ T]_{SS}$, where T and U are the linear transformations given above in (b) and S is the standard ordered basis for $\mathbb{R}^3$.

- (i) Deduce that $[U \circ T]_{SS} = [U]_{SS}[T]_{SS}$.
- (ii) Find $\text{Ker}(U \circ T)$.
- (iii) What is $\text{Rank}(U \circ T)$?

[8 Marks]

3.

(a) Consider the system of linear equations in the form $Ax = b$, where

$$A = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4, \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- (i) Find the condition on b_1, b_2 , and b_3 in order to have a solution for $Ax = b$ by applying elementary row operations to an appropriate matrix.
- (ii) Describe the column space of A and find a basis for it.
- (iii) Describe the Null space of A and find a basis for it.
- (iv) Describe the complete solution set of $Ax = b$, when $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

[15 Marks]

(b) Find the inverse of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ using Gauss-Jordan method.

[10 Marks]

4.

(a) Let $A = \begin{bmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{i1} & \cdots & a_{ii} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nn} \end{bmatrix}$ be a real square matrix of order n . Write down

the cofactor formula for $\det(A)$ with respect to the i^{th} row of A .

Suppose that D_n is the determinant of the 1, 1, 1 tridiagonal matrix

$$A_n = \begin{bmatrix} 1 & 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \text{ of order } n \text{ with}$$

$$A_1 = [1], A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, A_4 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \text{ and so on.}$$

- (i) Find D_1, D_2, D_3 , and D_4 .
- (ii) Use the cofactor formula for determinant along the first row to show that $D_n = D_{n-1} - D_{n-2}$ for $n \geq 2$.
- (iii) By finding the values of D_5, D_6, D_7 , and D_8 or otherwise, find D_{31} .

[14 Marks]

- (b) The vectors $a_1 = (1, 1, 1, 1)$ and $a_2 = (-2, -1, 0, 2)$ span a plane (two-dimensional subspace) in $\mathbb{R}^4$. Find the projection matrix P onto the plane, and find the projection of $b = (4, 3, 1, 0)$ onto the plane.

[11 Marks]